DHAYALAN K

DevOps Engineer | Cloud Infrastructure, Ansible, Docker, Amazon ECS & Linux Systems

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IMPACT AT A GLANCE

100% Idempotent Ansible Automation multi-node Linux fleet provisioning with UFW firewall & SSH hardening | **AWS ECS Knowledge Badge** Credly Verified for Amazon ECS Fargate, task definitions & cluster scaling | **AWS CDK & CloudFormation** declarative Infrastructure as Code CI/CD pipelines with managed deployment stages | **P2P Systems Daemon (ONey)** sub-20ms remote system management client with encrypted UDP tunnel

DevOps & Cloud Systems Engineer with practical experience in Infrastructure as Code (IaC), multi-node Linux server automation, container orchestration, and continuous delivery pipelines. Hands-on expertise managing production deployments using Ansible playbooks, Docker container runtimes, AWS ECS Fargate, AWS CloudFormation, and AWS CDK. Combines software engineering rigor in Python and Bash with deep Linux kernel and networking knowledge, having built custom remote access clients (ONey) and socket-level protocols (Z-Protocol).

CLOUD & INFRASTRUCTURE AUTOMATION

Ansible Fleet Automation & Server Hardening

- Authored modular Ansible automation playbooks for rapid, idempotent multi-node Linux server provisioning and base environment bootstrapping.
- Automated security hardening: enforcing UFW firewall default-deny incoming rules, allowing specific service ports (22, 80, 443), and disabling root password SSH logins via automated config validation.
- Structured playbooks into isolated roles with variable hierarchies and notified systemd handlers, eliminating configuration drift across target nodes.

AWS Container Orchestration & CI/CD Pipelines

- Engineered containerized workloads on Amazon ECS using Fargate launch types, task definitions, and auto-scaling rules (earned official Credly ECS Knowledge Badge).
- Constructed automated CI/CD deployment pipelines using AWS CDK with managed deployment stages, automating stack synthesis and rollouts.
- Authored declarative CloudFormation stacks for multi-tier resource orchestration, parameter isolation, and drift detection.
- Architected **ONey Remote System Control**: low-latency remote management client with encrypted P2P UDP hole punching, packaged into a supervised Linux `systemd` daemon.

TECHNICAL SKILLS

Infrastructure as Code (IaC): Ansible (Playbooks, Roles, Inventory, Handlers), AWS CloudFormation, AWS CDK, Bash Scripting, Cron Automation

Containers & Orchestration: Docker, Docker Compose, Amazon ECS (Task Definitions, Fargate, Cluster Scaling, Service Discovery), Containerd

Cloud Platforms (AWS): Amazon ECS, AWS CDK, CloudFormation, EC2, S3, IAM, VPC, CloudWatch, Application Load Balancers

Linux Administration & Security: Ubuntu, Debian, Systemd Service Configuration, UFW Firewall Security, SSH Hardening, Raw Sockets, TCP/UDP, DNS, Caddy/Nginx

CI/CD & Observability: GitHub Actions, Automated Build/Test/Lint Pipelines, Prometheus Node Exporter, Telemetry Streaming, Zero-Downtime Rollouts

PROFESSIONAL EXPERIENCE

DevOps & Cloud Systems Developer, Independent Systems & Open Source **2025 to Present, Coimbatore**

- Containerized production workloads on Amazon ECS with Fargate and auto scaling, earning the AWS ECS Knowledge Badge.
- Engineered automated CI/CD deployment pipelines using AWS CDK and declarative CloudFormation stacks.
- Created modular Ansible automation playbooks for rapid Linux server provisioning, SSH security, and UFW firewall enforcement.

Systems & Automation Engineer, Systems Engineering Lab **2024 to Present, Coimbatore**

- Architecting ONey — low-latency remote system management client with encrypted P2P tunneling and direct input synchronization.
- Automated filesystem maintenance, directory classification, and background maintenance daemons using Python and Bash.

KEY PROJECTS & CODEBASES

Ansible Fleet Provisioning & Linux Server Hardening github.com/DHAYALANKANIAPPAN/Ansible

- Built playbook-driven multi-node Linux provisioning and security hardening with 100% idempotent role execution.
- Automated UFW firewall policy enforcement and SSH security hardening. Stack: Ansible, YAML, Linux, Bash, UFW.

ONey Remote Systems Daemon & P2P Tunneling github.com/DHAYALANKANIAPPAN/PROJECTS

- Developed low-latency remote system management client with encrypted P2P UDP tunneling for host administration without relays.
- Packaged agent into supervised Linux systemd unit with automated restart policies. Stack: Python, UDP Sockets, Cryptography.

Z-Protocol Wire Transport Engine github.com/DHAYALANKANIAPPAN/Z-Protocol

- Layer-4 binary transport protocol on raw Python sockets with custom 16-byte framing, 3-way handshake state machine, and CRC32 verification.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (Cyber Security), Pursuing

Sri Shakthi Institute of Engineering and Technology, Coimbatore (2024 – 2028) | Grade Point Average: 8.21 out of 10.0.
Available for full-time employment & internships.

VERIFIED CERTIFICATIONS, AWARDS & ACTIVITIES

- Amazon ECS Knowledge Badge Assessment — Amazon Web Services (AWS) · Credly Official Badge.
- Deploying a CI/CD Pipeline with a Managed Deployment Stage using AWS CDK — AWS Skill Builder Official Lab.
- Using AWS CloudFormation for Automation — AWS Skill Builder Official Lab.
- Cloud Game Development Knowledge Badge — Amazon Web Services (AWS) · Credly Official Badge.
- 10+ AWS Skill Builder Deep-Dive Modules: ECS Security, Networking, Observability, Auto Scaling, Deployments & Workloads.